

7.7 Solution: (a) The wave equation for the electric field is

$$(\nabla^2 + \mu\epsilon\omega^2) \mathbf{E} = 0.$$

For $\mathbf{E} = \epsilon E(x)e^{i(kz-\omega t)}$, the equation reduces to

$$\frac{d^2 E(x)}{dx^2} + (\mu\epsilon\omega^2 - k^2)E(x) = 0.$$

Let $\kappa^2 = \mu\epsilon\omega^2 - k^2$, then the solution to the above equation is

$$E(x) = A(\kappa)e^{\pm i\kappa x}.$$

If k has some distribution near $\sqrt{\mu\epsilon}\omega$, then the solution should be the linear superposition, and to the lowest order approximation, we can take $k = \sqrt{\mu\epsilon}\omega$, *i.e.*,

$$E(x) = \int A(\kappa)e^{i\kappa x} d\kappa,$$

and

$$\mathbf{E} = \epsilon \int d\kappa A(\kappa)e^{i\kappa x + ikz - i\omega t}.$$

We can see that, for mode κ , the direction of propagation has a small deviation from the geometric optical path, with an angle of

$$\tan(\delta i_\kappa) = \frac{\kappa}{k},$$

or, $\delta i_\kappa = \kappa/k$, for small δi_κ .

(b) For the incident wave, we can write the electric field as

$$\mathbf{E}(\mathbf{x}, t) = \epsilon \int d\kappa A(\kappa)e^{i\kappa x_\perp + ikx_\parallel - i\omega t},$$

where $x_\parallel$ and $x_\perp$ are the components of $\mathbf{x}$ along the incident direction and the transverse direction, respectively. After reflection, k and κ will remain the same, but the projection of $\mathbf{x}$ onto the incident and transverse direction will be different, denoted as $x''_\parallel$ and $x''_\perp$. For mode κ , the totally reflected field also obtains a phase shift, $\phi(i + \delta i_\kappa, i_0)$, where $i + \delta i_\kappa$ is the incident angle of mode κ . Thus, the reflected field becomes

$$\mathbf{E}''(\mathbf{x}, t) = \epsilon'' \int d\kappa A(\kappa)e^{i\kappa x''_\perp + ikx''_\parallel - i\omega t + i\phi(i + \delta i_\kappa, i_0)}.$$

Expand the phase around i , we have

$$\begin{aligned} \mathbf{E}''(\mathbf{x}, t) &= \epsilon'' \int d\kappa A(\kappa) \exp\left(i\kappa x''_\perp + i\frac{d\phi(i, i_0)}{di}\delta i_\kappa\right) e^{ikx''_\parallel - i\omega t + i\phi(i, i_0)} \\ &= \epsilon'' \int d\kappa A(\kappa) \exp\left(i\kappa \left[x''_\perp + \frac{1}{k}\frac{d\phi(i, i_0)}{di}\right]\right) e^{ikx''_\parallel - i\omega t + i\phi(i, i_0)}. \end{aligned}$$

Recognize that

$$\int d\kappa A(\kappa) \exp\left(i\kappa \left[x''_\perp + \frac{1}{k}\frac{d\phi(i, i_0)}{di}\right]\right) = E\left(x''_\perp + \frac{1}{k}\frac{d\phi(i, i_0)}{di}\right) = E(x''_\perp - \delta x),$$

where

$$\delta x = -\frac{1}{k} \frac{d\phi(i, i_0)}{di},$$

we can write the reflected field as

$$\mathbf{E}''(\mathbf{x}, t) = \epsilon'' E(x''_{\perp} - \delta x) e^{ikx''_{\parallel} - i\omega t + i\phi(i, i_0)}.$$

(c) For $\mathbf{E}$ perpendicular to the incident plane, when there is total reflectoin,

$$i > i_0 = \arcsin\left(\frac{n'}{n}\right),$$

Eq. (7.39) for the reflected field becomes

$$\frac{E''_0}{E_0} = \frac{\cos i - i\sqrt{\sin^2 i - \sin^2 i_0}}{\cos i + i\sqrt{\sin^2 i - \sin^2 i_0}},$$

assuming $\mu = \mu'$, *i.e.*, reflection from a nonmagnetic medium. Then,

$$\phi_{\perp}(i, i_0) = -2 \arctan\left(\frac{\sqrt{\sin^2 i - \sin^2 i_0}}{\cos i}\right).$$

Simple differentiation shows that

$$\frac{d\phi_{\perp}(i, i_0)}{di} = -\frac{2 \sin i}{\sqrt{\sin^2 i - \sin^2 i_0}}.$$

Also, $k = 2\pi/\lambda$, and

$$D_{\perp} = -\frac{1}{k} \frac{d\phi_{\perp}(i, i_0)}{di} = \frac{\lambda}{\pi} \frac{\sin i}{\sqrt{\sin^2 i - \sin^2 i_0}}.$$

Similarly, for $\mathbf{E}$ parallel to the incident plane,

$$\phi_{\parallel}(i, i_0) = -2 \arctan\left(\frac{\sqrt{\sin^2 i - \sin^2 i_0}}{\sin^2 i_0 \cos i}\right),$$

and it can be directly verified that

$$D_{\parallel} = D_{\perp} \frac{\sin^2 i_0}{\sqrt{\sin^2 i - \cos^2 i \cdot \sin^2 i_0}}.$$